

From Criteria to Ranking: Targeting-Aware Tripartite Graph Learning for Multi-Criteria Recommendation

Zhenhua Meng*
Inner Mongolia University
Hohhot, China
zhmeng@imu.edu.cn

Fanshen Meng
Beijing University of Posts and Telecommunications
Beijing, China
mengfanshen@bupt.edu.cn

Abstract

Multi-criteria recommender systems (MCRSs) are becoming increasingly important in the Web ecosystem, where platforms such as e-commerce sites and review portals allow users to evaluate items from multiple perspectives. By leveraging criterion-level ratings rather than relying solely on overall scores, MCRSs can enhance personalization and more accurately capture user preferences. However, existing multi-criteria recommendation methods often fail to explicitly model user-specific preferences across criteria or to incorporate item-criterion signals into representation learning. To solve these problems, we propose **TaTriGR**, a Targeting-Aware Tripartite Graph Recommender, which jointly models user-item-criterion interactions in a unified tripartite graph and integrates user-specific criterion weights through targeting mechanisms. Specifically, TaTriGR encodes user-criterion targeting through personalized weights, while incorporating item-criterion performance as an additional channel to enrich item semantics. A lightweight propagation mechanism then diffuses information across the tripartite structure, and a formal score decomposition shows that predictions satisfy fundamental multi-criteria decision making (MCDM) properties. To further reinforce targeting, TaTriGR introduces two auxiliary objectives: Ideal Point Distillation and Lexicographic Consistency, which encourage criterion-consistent user representations and rankings. Extensive experiments on three real-world datasets demonstrate the effectiveness of TaTriGR, showing relative gains up to 16.1% over the best baseline models. Our implementations are available at <https://github.com/nunu1995/TaTriGR>.

CCS Concepts

• Information systems → Recommender systems.

Keywords

Multi-Criteria Recommender Systems, Graph Neural Networks, Multi-Criteria Decision Making, Preference Learning

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*Corresponding author.



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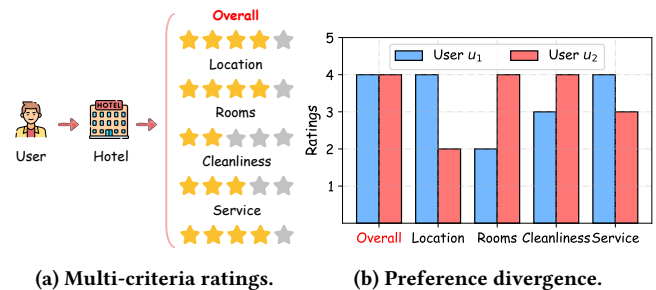


Figure 1: Illustration of multi-criteria recommendation: (a) users provide detailed ratings across multiple criteria in addition to an overall rating; (b) users with the same overall rating may exhibit distinct criterion-level preferences.

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1 Introduction

Multi-criteria recommender systems (MCRSs) extend conventional recommendation by enabling users to evaluate items from multiple dimensions rather than relying solely on an overall rating [19, 21, 28, 32]. Such information reflects the diversity of user preferences, thereby supporting more accurate recommendations and offering clearer explanations for item selection. MCRSs have been widely applied in Web-based platforms such as e-commerce, travel, hotels and movies [3, 26, 29, 39]. For example, as illustrated in Figure 1a, when evaluating hotels, users provide not only an overall rating but also detailed feedback on aspects such as *location*, *rooms*, *cleanliness*, and *service*. This often results in cases where two users assign the same overall rating but differ significantly in their criterion evaluations. As shown in Figure 1b, user u_1 values *location* and *service* but is dissatisfied with *rooms*, whereas user u_2 emphasizes *rooms* and *cleanliness* while remaining less concerned about *location*. Although both users give an overall rating of 4, the information conveyed by their multi-criteria evaluations is highly divergent. This indicates that, compared with MCRSs, single-criterion recommendation systems may overlook user preference differences and lead to biased results [18]. In contrast, multi-criteria ratings can provide a richer and more faithful representation of user decision behaviors [11, 31, 37].

Research on MCRSs has primarily focused on how to effectively model criterion information [2, 5, 9, 35]. Early studies often extended collaborative filtering (CF) or matrix factorization (MF) in multi-criteria scenarios by aggregating criterion-level ratings to

predict the overall rating, employing approaches such as multilinear matrix factorization, stacked autoencoders, variational autoencoders, and deep autoencoders [21, 22, 34, 37]. More recently, graph-based recommendation methods have become increasingly prominent [4, 38]. By representing multi-criteria data as graph structures, many methods leverage high-order connectivity between users and items, with several further incorporating criterion-aware message passing or graph filtering to capture inter-criterion relations [31, 32]. In addition, hyperbolic manifolds and causal inference have also been introduced to enrich the modeling capacity of MCRSs [11, 12]. Despite these advancements, existing most models still lack a clear mechanism to characterize how users target specific criteria during the decision-making process. In other words, while they improve representational power, they do not directly address the criteria-oriented prioritization in user modeling.

From a decision-making perspective, multi-criteria recommendation inherently involves ranking items based on multiple criteria [1]. In multi-criteria decision-making (MCDM), alternatives are typically evaluated across multiple dimensions to identify the overall best option [15, 16, 23]. However, real-world decision-making often deviates from this assumption: **decision-makers often care less about the overall best alternatives and more about alternatives that perform best relative to a specific target** [17]. This target-oriented perspective naturally aligns with user behavior in MCRSs. For example, in Figure 1b, user u_1 values *location* and *service* more highly than other criteria. If these criteria are treated as the target, the recommender system can recommend hotels that perform well in them rather than simply suggesting those with the highest overall ratings. Introducing targeting into MCRSs offers two key advantages: **i)** it provides a principled mechanism to identify the criteria most relevant to user preferences, **ii)** and it ensures that recommendations are consistent with the decision strategies users naturally adopt. However, criterion targeting also poses several challenges. **C1:** How to automatically identify which criteria are most important for a user's targeting? **C2:** How to design aggregation functions that are consistent with fundamental MCDM properties while remaining explainable? **C3:** How to effectively integrate target preferences with CF signals in a unified recommendation framework?

To address the above challenges, we propose a Targeting-Aware Tripartite Graph Recommender (**TaTriGR**), which explicitly models the user–item–criterion tripartite structure and integrates targeting for multi-criteria recommendation. First, we design novel user–criterion weighting strategies that quantify the relative contribution of each criterion to user preferences by analyzing interaction patterns, allowing the model to identify the dimensions emphasized during recommendation (**C1**). Second, we introduce targeting-based auxiliary terms, including Ideal Point Deviation (IPD), which aligns user embeddings with their weighted criterion representations, and Lexicographic Consistency (LC), which constrains ranking consistency by requiring items superior on emphasized criteria to be ranked higher than alternatives. These objectives regularize the relationship between user and criterion embeddings and reflect criterion-level prioritization in user representations (**C2**). Finally, we construct a user–item–criterion tripartite graph where users, items, and criteria are represented as nodes, and we develop a lightweight graph propagation mechanism for aggregating information

across user–item, user–criterion, and item–criterion edges. In this way, targeting signals and collaborative relations are jointly encoded within a shared embedding space (**C3**). Building upon this framework, the main contributions of this paper are summarized as follows:

- We introduce the concept of targeting into multi-criteria recommendation, bridging the gap between existing approaches that either aggregate criterion ratings or treat them independently, and real decision-making behaviors where users emphasize specific criteria when forming preferences.
- We propose TaTriGR, a targeting-aware tripartite graph framework that represents users, items, and criteria as nodes. It incorporates targeting-based auxiliary losses, including IPD and LC to capture criterion-level prioritization, while employing lightweight graph propagation to encode targeting signals with collaborative relations.
- We conduct extensive experiments on three public datasets, showing that TaTriGR outperforms state-of-the-art multi-criteria recommender models. Ablation studies further validate the effectiveness of both our targeting mechanisms and the tripartite graph design.

2 Preliminaries

Let $\mathcal{U} = u_1, \dots, u_{|\mathcal{U}|}$ and $\mathcal{I} = i_1, \dots, i_{|\mathcal{I}|}$ denote the user and item sets, respectively. For each user $u \in \mathcal{U}$, let $\mathcal{N}_u \subseteq \mathcal{I}$ be the set of items interacted with by u . In multi-criteria recommendation, each user–item interaction is associated not only with an overall rating but also with a set of criterion-level ratings that reflect different aspects of user evaluation. The collection of rating matrices is defined as $\mathcal{R} = \{\mathcal{R}_0, \mathcal{R}_1, \dots, \mathcal{R}_{|C|}\}$, where $\mathcal{R}_0 \in \mathbb{R}^{|\mathcal{U}| \times |\mathcal{I}|}$ corresponds to the overall ratings, $\mathcal{R}_c \in \mathbb{R}^{|\mathcal{U}| \times |\mathcal{I}|}$ represents the ratings on the c -th criterion, and $C = \{c_1, c_2, \dots, c_{|C|}\}$ denotes the set of criteria.

The objective of multi-criteria recommendation is to learn a scoring function $y_\theta : \mathcal{U} \times \mathcal{I} \rightarrow \mathbb{R}$, that predicts the overall preference of user u for an item i by leveraging information from the overall rating matrix $\mathcal{R}_0$ and the criterion matrices $\{\mathcal{R}_c\}_{c=1}^{|C|}$. Based on this scoring function, the system generates a top- K recommendation list for each user:

$$\mathcal{L}_u^K = \text{Top}K_{i \in \mathcal{I} \setminus \mathcal{N}_u} y_\theta(u, i), \quad (1)$$

which contains the K items that user u is most likely to prefer.

3 Methodology

In this section, we present the proposed TaTriGR, and its overall framework is shown in Figure 2.

3.1 Tripartite Graph Construction

3.1.1 Node Definition. To integrate multi-criteria signals into the recommendation process, it is essential to construct a structure that encodes heterogeneous entities and their relations. In this work, we formalize the recommendation scenario as a tripartite graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \mathcal{U} \cup \mathcal{I} \cup \mathcal{C}$ denotes the set of nodes, consisting of users $\mathcal{U} = \{u_1, u_2, \dots, u_{|\mathcal{U}|}\}$, items $\mathcal{I} = \{i_1, i_2, \dots, i_{|\mathcal{I}|}\}$, and criteria $\mathcal{C} = \{c_1, c_2, \dots, c_{|\mathcal{C}|}\}$. Each type of node plays a distinct semantic role, where user nodes act as preference sources, item nodes represent recommendation targets, and criterion nodes function as

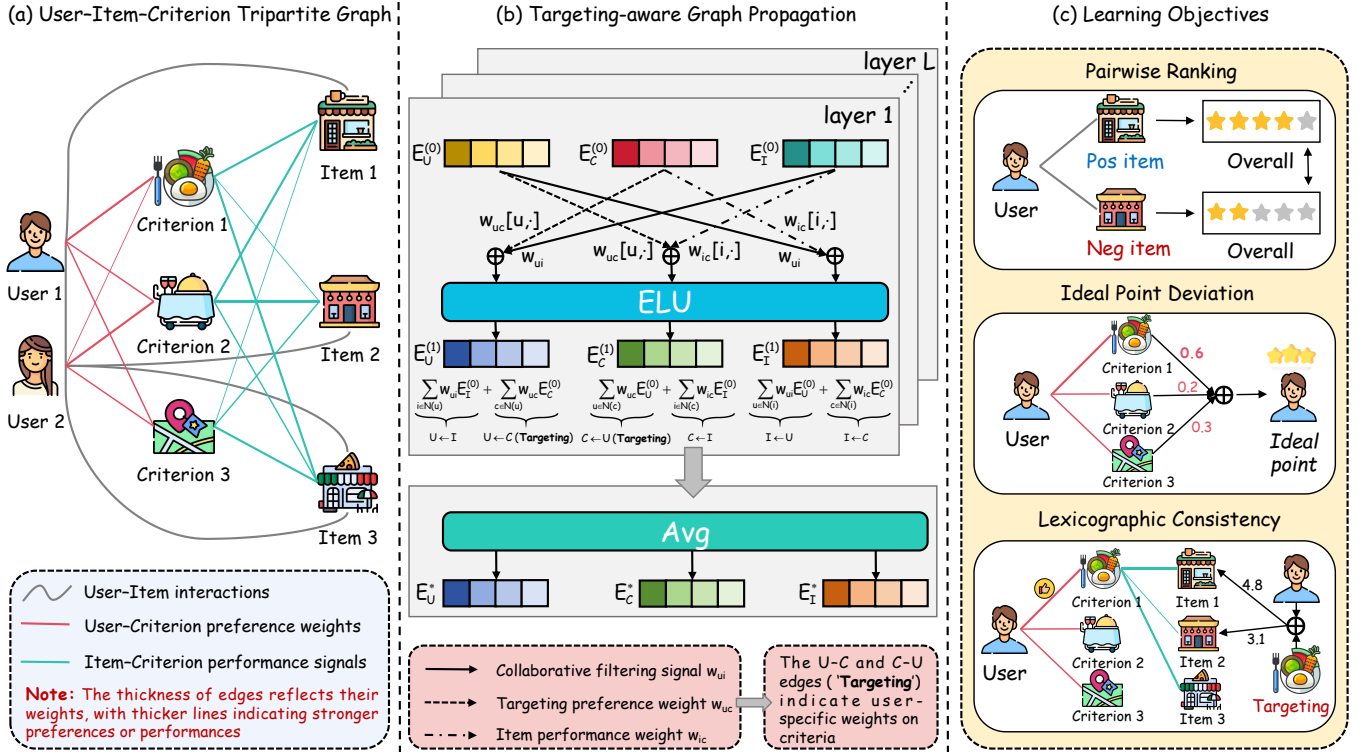


Figure 2: Overview of TaTriGR. (a) Tripartite graph construction: users, items, and criteria are represented as nodes, with edges encoding user-item, user-criterion, and item-criterion relations. **(b) Graph propagation:** a lightweight layer-wise aggregation over U-I, U-C, and I-C connections. **(c) Learning objectives:** jointly optimization of BPR loss for recommendation ranking, IPD loss for aligning users embeddings with weighted criterion profiles, and LC loss for enforcing criterion-level prioritization.

mediating factors that bridge user preferences with item characteristics. The edge set is defined as $\mathcal{E} = \mathcal{E}_{UI} \cup \mathcal{E}_{UC} \cup \mathcal{E}_{IC}$, which captures three categories of relations correspond to user-item, user-criterion, and item-criterion interactions.

3.1.2 Edge Formation. Let $\Omega \subseteq \mathcal{U} \times \mathcal{I}$ be the index set of observed user-item interactions. For each $(u, i) \in \Omega$, we use $r_{ui} = (\mathcal{R}_0)_{ui}$ to represent the overall rating of user u on item i , and $r_{ui}^c = (\mathcal{R}_c)_{ui}$ to denote the rating with respect to criterion c .

User-item edges $\mathcal{E}_{UI}$. An edge $(u, i) \in \mathcal{E}_{UI}$ is formed if user u has provided an overall rating for item i . The corresponding edge weight is defined as the normalized overall rating:

$$w_{ui} = \frac{r_{ui} - r_{\min}}{r_{\max} - r_{\min}}, \quad (2)$$

where $r_{\min}$ and $r_{\max}$ are the minimum and maximum values of the rating scale, respectively.

User-criterion edges $\mathcal{E}_{UC}$. An edge $(u, c) \in \mathcal{E}_{UC}$ represents the preference strength of user u toward criterion c . The overall mean rating across all criteria for user u is defined as

$$\bar{r}_u = \frac{1}{|C|} \sum_{c \in C} \left(\frac{1}{|\mathcal{N}_u^c|} \sum_{i \in \mathcal{N}_u^c} r_{ui}^c \right), \quad (3)$$

where $\mathcal{N}_u^c$ is the set of items that user u has rated under criterion c . To alleviate sparsity, we introduce a centered preference score:

$$p_{uc} = \frac{|\mathcal{N}_u^c|}{1 + s_{uc}} (\bar{r}_{uc} - \bar{r}_u), \quad (4)$$

where $s_{uc}^2 = \text{Var}_{i \in \mathcal{N}_u^c} [r_{ui}^c]$ denotes the variance of user u 's ratings under criterion c , and $\bar{r}_{uc}$ is the average rating of user u on criterion c . Finally, the criterion-specific preference distribution is normalized via Softmax function:

$$w_{uc} = \frac{\exp(\alpha p_{uc})}{\sum_{c' \in C} \exp(\alpha p_{uc'})}, \quad (5)$$

where α is a scaling factor controlling sharpness of the distribution.

Item-criterion edges $\mathcal{E}_{IC}$. An edge $(i, c) \in \mathcal{E}_{IC}$ represents the performance of item i with respect to criterion c . The raw score is calculated as the average rating:

$$\bar{r}_{ic} = \frac{1}{|\mathcal{N}_i^c|} \sum_{u \in \mathcal{N}_i^c} r_{ui}^c, \quad (6)$$

where $\mathcal{N}_i^c$ is the set of users who have rated item i under criterion c . The global mean score on criterion c is defined as

$$\bar{r}_c = \frac{1}{\sum_i |\mathcal{N}_i^c|} \sum_i \sum_{u \in \mathcal{N}_i^c} r_{ui}^c. \quad (7)$$

To stabilize the estimates for items with only a few ratings, we apply Bayesian smoothing:

$$\tilde{r}_{ic} = \frac{|\mathcal{N}_i^c| \cdot \bar{r}_{ic} + \beta \cdot \bar{r}_c}{|\mathcal{N}_i^c| + \beta}, \quad (8)$$

where β is a shrinkage parameter controlling the shrinkage strength. Finally, the smoothed score is normalized into $[0, 1]$:

$$w_{ic} = \frac{\tilde{r}_{ic} - \min_{i'} \tilde{r}_{i'c}}{\max_{i'} \tilde{r}_{i'c} - \min_{i'} \tilde{r}_{i'c}}. \quad (9)$$

3.1.3 Graph Representation. With the node types and edge weights defined above, the tripartite graph can be expressed as

$$\mathcal{G} = (\mathcal{U}, \mathcal{I}, \mathcal{C}, \mathcal{E}_{\mathcal{UI}}, \mathcal{E}_{\mathcal{UC}}, \mathcal{E}_{\mathcal{IC}}).$$

The adjacency matrix $\mathbf{A} \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$ of $\mathcal{G}$ incorporates the weighted submatrices $\mathbf{W}_{\mathcal{UI}}$, $\mathbf{W}_{\mathcal{UC}}$, and $\mathbf{W}_{\mathcal{IC}}$, corresponding to the three edge categories. Specifically, the entries are defined as

$$\mathbf{A}[u, i] = \mathbf{W}_{\mathcal{UI}}[u, i], \quad \mathbf{A}[u, c] = \mathbf{W}_{\mathcal{UC}}[u, c], \quad \mathbf{A}[i, c] = \mathbf{W}_{\mathcal{IC}}[i, c],$$

for $u \in \mathcal{U}$, $i \in \mathcal{I}$, and $c \in \mathcal{C}$, while all other entries are set to zero.

3.2 Graph Propagation with Targeting

3.2.1 Propagation Rule. Let $\mathbf{E}^{(l)} = [\mathbf{E}_{\mathcal{U}}^{(l)}; \mathbf{E}_{\mathcal{I}}^{(l)}; \mathbf{E}_{\mathcal{C}}^{(l)}]$ denote the embeddings of users, items, and criteria at the l -th layer, where $\mathbf{E}^{(0)}$ is the initial embeddings. To ensure stable message passing in the tripartite graph, we adopt symmetric normalization of the adjacency matrices. For each relation pair $(X, Y) \in \{(\mathcal{U}, \mathcal{I}), (\mathcal{U}, \mathcal{C}), (\mathcal{I}, \mathcal{C})\}$, the normalized adjacency matrix is defined as

$$\hat{\mathbf{A}}_{XY} = \mathbf{D}_X^{-\frac{1}{2}} \mathbf{A}_{XY} \mathbf{D}_Y^{-\frac{1}{2}}, \quad (10)$$

where $\mathbf{D}_X$ and $\mathbf{D}_Y$ are the diagonal degree matrices of $\mathbf{A}_{XY}$. At each layer, embeddings are updated by aggregating messages from connected neighbors. For instance, the update of a user embedding is formulated as

$$\mathbf{e}_u^{(l+1)} = \text{ELU}\left(\sum_{i \in \mathcal{N}_u} \hat{\mathbf{A}}_{ui} \mathbf{e}_i^{(l)} + \sum_{c \in \mathcal{C}} \hat{\mathbf{A}}_{uc} \mathbf{e}_c^{(l)}\right), \quad (11)$$

where $\text{ELU}(\cdot)$ is the Exponential Linear Unit activation function, and the weights $\hat{\mathbf{A}}_{uc} \propto w_{uc}$ embed user-specific targeting preferences derived from criterion distributions.

3.2.2 Scoring Decomposition. The predicted score of user u for item i is defined as the inner product $\hat{y}_{ui} = \langle \mathbf{e}_u^*, \mathbf{e}_i^* \rangle$, where $\mathbf{e}_u^*$ and $\mathbf{e}_i^*$ are the final representations of u and i , obtained by layer-wise aggregation:

$$\mathbf{e}_x^* = \frac{1}{L+1} \sum_{l=0}^L \mathbf{e}_x^{(l)}. \quad (12)$$

As outlined in the previous section, $\hat{y}_{ui}$ contains a component that aggregates criterion-specific signals with w_{uc} as coefficients. Let

$$\phi_c(i) = \langle \mathbf{e}_c^{(0)}, \sum_{l=0}^L \gamma_l \mathbf{e}_i^{(l)} \rangle, \quad (13)$$

where $\{\gamma_l\}$ are nonnegative coefficients induced by the propagation. The final score can then be decomposed as

$$\hat{y}_{ui} = \sum_{c \in \mathcal{C}} w_{uc} \phi_c(i) + \psi(u, i), \quad (14)$$

where the first term reflects a criterion-targeted aggregation of item performances according to user preferences, and the residual term $\psi(u, i)$ collects contributions from non-criterion paths.

3.2.3 Targeting Properties. Based on the score decomposition in Eq. (14), we can formally establish several fundamental properties that the targeting-aware propagation mechanism satisfies:

- **P1 (Neutrality):** Rankings depend only on $(\phi_c(i))_c$ and $\{w_{uc}\}$, but not on item identities themselves.
- **P2 (Contribution):** If $w_{uc} = 0$, the contribution of criterion c vanishes, so it exerts no influence on the prediction.
- **P3 (Responsiveness):** If item i improves on criterion c such that $\phi_c(i)$ strictly increases and $w_{uc} > 0$, then $\hat{y}_{ui}$ strictly increases, and the ranking is adjusted accordingly.
- **P4 (Contraction):** Since the final score is induced by point-wise aggregation, restricting the candidate set from $\mathcal{O}$ to a subset $\mathcal{O}' \subseteq \mathcal{O}$ preserves the relative order of items in $\mathcal{O}'$.

THEOREM 3.1. Fix a user u . Suppose that i) the $\{w_{uc}\}_{c \in \mathcal{C}}$ form a probability distribution with $w_{uc} > 0$ and $\sum_c w_{uc} = 1$; ii) each $\phi_c(i)$ in Eq. (14) is a nondecreasing function of the item-criterion strength (e.g., the smoothed score $\tilde{r}_{ic}$); and iii) the residual satisfies $\psi(u, i) = \psi_u$ for all $i \in \mathcal{O}$, i.e., item-independent within the candidate set. Then the induced preference can be equivalently written as

$$\arg \max_{i \in \mathcal{O}} \hat{y}_{ui} = \arg \max_{i \in \mathcal{O}} f^{(u)}((\phi_c(i))_{c \in \mathcal{C}}), \quad (15)$$

$$f^{(u)}(v_1, \dots, v_{|\mathcal{C}|}) = \sum_c w_{uc} v_c, \quad (16)$$

for any finite candidate set $\mathcal{O} \subseteq \mathcal{I}$. Therefore, $\hat{y}_{ui}$ satisfies properties P1–P4. In practice, although $\psi(u, i)$ may vary across items, our experiments (Section 4.3) show that the criterion-weighted component dominates the prediction, so these properties hold approximately and guide the model's preference behavior.

3.3 Optimization

3.3.1 Overall Ranking. We adopt the Bayesian Personalized Ranking (BPR) loss [33] to optimize the relative order between observed and unobserved items for each user, which is defined as

$$\mathcal{L}_{\text{BPR}} = - \sum_{(u, i^+, i^-) \in \mathcal{D}} \ln \sigma(\hat{y}_{ui^+} - \hat{y}_{ui^-}), \quad (17)$$

where $\mathcal{D}$ denotes the set of training triplets (u, i^+, i^-) , i^+ is an item with which user u has interacted, and i^- is a randomly sampled unobserved item.

3.3.2 Ideal Point Distillation (IPD). To ensure that user embeddings faithfully reflect criterion-level preferences, we introduce an ideal point distillation regularizer. The intuition is that each user embedding $\mathbf{e}_u^*$ should remain close to the user's *ideal point*, defined as the criterion-weighted aggregation of criterion embeddings. Formally, the IPD loss is given by

$$\mathcal{L}_{\text{IPD}} = \sum_{u \in \mathcal{U}} \|\mathbf{e}_u^* - \sum_{c \in \mathcal{C}} w_{uc} \mathbf{e}_c^*\|_2^2. \quad (18)$$

3.3.3 Lexicographic Consistency (LC). In practical decision-making, users often emphasize their dominant criteria when evaluating alternatives. To capture this behavior, we design a LC constraint: if item i outperforms item j on user u 's top-weighted criterion

Table 1: Statistics of the datasets.

Dataset	#Users	#Items	#Interactions	#Criteria	Density
TA	4,264	6,274	34,383	7	1.30e-3
YM	1,826	1,469	50,608	4	1.89e-2
RB	4,020	3,380	161,276	4	1.19e-2

$c^* = \arg \max_c w_{uc}$, then i should also be ranked higher in the final prediction. The LC loss is defined as

$$\mathcal{L}_{LC} = - \sum_{(u,i,j)} \log \sigma((r_{ui}^{c^*} - r_{uj}^{c^*}) \cdot (\hat{y}_{ui} - \hat{y}_{uj})), \quad (19)$$

where $r_{ui}^{c^*}$ represents user u 's rating of item i under the dominant criterion c^* .

3.3.4 Final Objective. The overall learning objective combines BPR loss with the two auxiliary regularization components:

$$\mathcal{L} = \mathcal{L}_{BPR} + \lambda_1 \mathcal{L}_{IPD} + \lambda_2 \mathcal{L}_{LC} + \lambda \|\Theta\|_2^2, \quad (20)$$

where λ_1 and λ_2 control the contributions of the IPD and LC losses, and λ regulates the strength of L_2 regularization on parameters Θ .

4 Experiments

In this section, we conduct extensive experiments on benchmark datasets to address the following research questions:

- **RQ1:** How does TaTriGR perform compared to various multi-criteria recommendation methods?
- **RQ2:** How does each component of TaTriGR contribute to its overall performance?
- **RQ3:** How does targeting influence the recommendation performance and ranking behavior?
- **RQ4:** How do different hyperparameter configurations affect the performance of TaTriGR?

4.1 Experimental Settings

4.1.1 Datasets. We evaluate TaTriGR on three widely used datasets for multi-criteria recommendation: **TripAdvisor (TA)**, **Yahoo! Movies (YM)**, and **RateBeer (RB)** [7, 27, 30]. These datasets cover different domains and provide both overall ratings and multi-criteria ratings. To ensure data reliability and sufficient user activity, we filter out users with fewer than five interactions. The key statistics of the datasets are summarized in Table 1.

4.1.2 Baselines. To validate the effectiveness of TaTriGR, we compare it with three categories of recommender models. **i) Multi-criteria Recommenders:** ExtendedSAE [37], LatentMC [21], UBM [46], DMCF [29], CFM [7], and AEMC [34]. **ii) Single-criterion Recommenders:** SpectralCF [45], NGCF [40], LightGCN [13], SGL [41], and AutoCF [42]. **iii) Graph-based Multi-criteria Recommenders:** LightGCN_{MC}, SGL_{MC}, CPA-LGC [31], CA-GF [32], HMCR [11], and MCCR [12]. Here, LightGCN_{MC} and SGL_{MC} are variants of LightGCN [13] and SGL [41] for the multi-criteria setting.

4.1.3 Evaluation Metrics. To evaluate the top- K multi-criteria recommendation performance, we adopt three widely used ranking metrics [12–14, 25, 31]: **Hit Ratio (H@K)**, **Recall (R@K)**, and **Normalized Discounted Cumulative Gain (N@K)**, with $K \in \{20, 50\}$.

4.1.4 Implementation Details. We partition each dataset into training (70%), validation (10%), and test (20%) sets based on user interactions. For TaTriGR, the embedding dimension is set to $d = 64$, and the number of graph propagation layers is fixed to $L = 2$. Model parameters are initialized using the Xavier scheme [10], and optimization is performed with Adam [20] under mini-batch training of size 1024. The learning rate and regularization weight are tuned from $1e-4$ to $1e-1$. The parameters λ_1 and λ_2 are searched in $\{1e-4, 1e-3, 1e-2, 1e-1, 1\}$. The sharpness parameter α is tuned in 1 to 5, and the smoothing parameter β is selected from $\{1, 5, 10, 15, 20\}$. For all baselines, we adopt the recommended hyperparameter configurations reported in their original papers. All experiments are conducted on an NVIDIA GeForce RTX 4090 GPU.

4.2 Performance Comparison (RQ1)

4.2.1 Comparisons w.r.t. Overall Performance. To evaluate the performance of TaTriGR, we compare it against the baselines described in Section 4.1.2. Table 2 reports the top-20 and top-50 results on the TA, YM, and RB datasets under three evaluation metrics, from which we derive the following observations:

- TaTriGR achieves the best performance across all datasets and metrics. For example, on TA it outperforms the strongest baseline MCCR by 6.59% in H@20 and 7.45% in N@20. On YM, it achieves an R@20 of 0.2355, corresponding to gains of 12.73% over MCCR and 15.48% over HMCR. On RB, it delivers the largest improvements in R@50, with relative gains of 78.79% over the best multi-criteria baseline, 41.01% over the best single-criterion baseline, and 14.28% over the best graph-based multi-criteria baseline. These results confirm that TaTriGR generalizes across domains with different sparsity and criterion characteristics.
- Multi-criteria recommenders based on CF or MF extensions provide only marginal benefits. On YM, TaTriGR achieves an N@50 of 0.3627, which is nearly $2.9\times$ higher than LatentMC. On RB, it doubles or even triples the performance of these baselines across most metrics. This confirms the necessity of modeling user–criterion targeting and item–criterion signals within a unified tripartite structure.
- Single-criterion recommenders achieve strong performance on YM and RB where overall ratings are denser, but fail to capture criterion-level heterogeneity in user preferences. TaTriGR consistently outperforms them, particularly on recall-oriented metrics. For example, on YM it raises R@50 from 0.2885 with SGL to 0.3600, and on RB it increases N@20 from 0.2892 with SGL to 0.4293. These demonstrate that incorporating criterion signals yields gains beyond what can be learned from overall ratings alone.
- Graph-based multi-criteria recommenders are the strongest baselines. Multi-criteria extensions clearly enhance recommendation performance, as LightGCN_{MC} surpasses LightGCN and SGL_{MC} improves over SGL, showing that criterion signals complement overall ratings. However, they lack user-specific targeting, which explains why TaTriGR achieves further gains. On TA, it improves R@20 from 0.0896 with MCCR to 0.1007, yielding a 12.39% gain. On YM, it raises N@20 from 0.2879 with MCCR to 0.2942 and improves recall

Table 2: The overall performance comparison between TaTriGR and baselines on TA, YM and RB datasets. The best and second-best results are highlighted in bold and underline, respectively. “Improv.” indicates the relative improvement of TaTriGR over the best-performing baseline model.

Dataset	TA						YM						RB					
Metrics	H@20	H@50	R@20	R@50	N@20	N@50	H@20	H@50	R@20	R@50	N@20	N@50	H@20	H@50	R@20	R@50	N@20	N@50
Multi-criteria Recommenders																		
ExtendedSAE	0.0205	0.0367	0.0125	0.0283	0.0051	0.0073	0.1490	0.2023	0.1271	0.1493	0.1437	0.1521	0.0303	0.0571	0.0223	0.0483	0.0181	0.0213
LatentMC	0.1063	0.1385	0.0723	0.0867	0.0593	0.0647	0.1214	0.2027	0.0913	0.1645	0.1185	0.1231	0.1357	0.2361	0.1131	0.2067	0.1939	0.2092
UBM	0.0757	0.0987	0.0583	0.0737	0.0391	0.0424	0.0455	0.0929	0.0417	0.0723	0.0289	0.0323	0.0561	0.1137	0.0551	0.0997	0.0523	0.0629
DMCF	0.0463	0.0853	0.0315	0.0485	0.0287	0.0325	0.1054	0.1437	0.0793	0.1281	0.0925	0.1039	0.1502	0.2129	0.1337	0.2213	0.1925	0.2087
CFM	0.0857	0.1427	0.0803	0.0927	0.0653	0.0716	0.1533	0.2161	0.1153	0.1950	0.0915	0.1027	0.2828	0.3037	0.2357	0.3419	0.1837	0.2017
AEMC	0.0497	0.0883	0.0311	0.0477	0.0270	0.0313	0.0867	0.1523	0.0635	0.1153	0.0751	0.0859	0.1257	0.2207	0.1073	0.1985	0.1837	0.2041
Single-criterion Recommenders																		
SpectralCF	0.0705	0.1051	0.0308	0.0547	0.0391	0.0509	0.4203	0.5807	0.1365	0.2352	0.1502	0.1764	0.6752	0.7253	0.2704	0.3907	0.2621	0.2955
NGCF	0.0944	0.1312	0.0468	0.0639	0.0385	0.0529	0.4354	0.6042	0.1425	0.2467	0.1583	0.1813	0.6881	0.7389	0.2784	0.4010	0.2680	0.2997
LightGCN	0.1165	0.1597	0.0608	0.0796	0.0596	0.0713	0.4972	0.7011	0.1611	0.2781	0.1765	0.1981	0.7120	0.7683	0.2927	0.4189	0.2786	0.3139
SGL	0.1235	0.1752	0.0649	0.0902	0.0651	0.0750	0.5286	0.7270	0.1723	0.2885	0.1845	0.2122	0.7336	0.7830	0.3038	0.4300	0.2892	0.3240
AutoCF	0.1269	0.1765	0.0667	0.0915	0.0682	0.0734	0.5157	0.7300	0.1669	0.2913	0.1831	0.2055	0.7264	0.7850	0.2992	0.4335	0.2841	0.3205
Graph-based Multi-criteria Recommenders																		
LightGCN _{MC}	0.1382	0.1887	0.0673	0.1008	0.0692	0.0819	0.5608	0.7524	0.1762	0.2906	0.1867	0.2189	0.7654	0.8697	0.3312	0.4685	0.3159	0.3578
SGL _{MC}	0.1426	0.1924	0.0708	0.1043	0.0712	0.0877	0.5759	0.7708	0.1824	0.3009	0.1955	0.2266	0.7728	0.8803	0.3387	0.4806	0.3264	0.3665
CPA-LGC	0.1468	0.1983	0.0732	0.1069	0.0827	0.0954	0.5821	0.7762	0.1868	0.3057	0.1981	0.2345	0.7906	0.8894	0.3442	0.4978	0.3358	0.3759
CA-GF	0.1509	0.2026	0.0764	0.1092	0.0859	0.0988	0.5967	0.7915	0.1926	0.3124	0.2055	0.2426	0.8023	0.9028	0.3515	0.5087	0.3427	0.3825
HMC	0.1617	0.2112	0.0822	0.1108	0.0897	<u>0.1045</u>	0.6220	0.7953	0.2041	0.3178	0.2535	0.2791	<u>0.8573</u>	<u>0.9410</u>	<u>0.3618</u>	<u>0.5349</u>	<u>0.4145</u>	<u>0.5198</u>
MCCR	0.1655	0.2158	0.0896	<u>0.1117</u>	<u>0.0913</u>	0.1036	<u>0.6415</u>	<u>0.8007</u>	<u>0.2089</u>	<u>0.3258</u>	<u>0.2879</u>	<u>0.3576</u>	<u>0.8526</u>	<u>0.9371</u>	<u>0.3584</u>	<u>0.5296</u>	<u>0.4102</u>	<u>0.5132</u>
TaTriGR	0.1764	0.2205	0.1007	0.1297	0.0981	0.1072	0.6566	0.8092	0.2355	0.3600	0.2942	0.3627	0.8692	0.9435	0.4188	0.6113	0.4293	0.5254
Improv.	+6.59%	+2.18%	+12.39%	+16.11%	+7.45%	+2.58%	+2.35%	+1.06%	+12.73%	+10.50%	+2.19%	+1.43%	+1.39%	+0.27%	+15.75%	+14.28%	+3.57%	+1.08%

by over 12%. Notably, HMC only outperforms MCCR on RB, suggesting that its hierarchical design works better suited to sparse and noisy criteria but less stable on denser datasets.

4.2.2 Comparisons w.r.t. Data Sparsity. To evaluate the performance of TaTriGR under different sparsity levels, we follow the strategy in [43, 44] and divide users into four groups based on their interaction scale. As shown in Figure 3, the sparsest group accounts for over 50% of users in both YM and RB datasets. From the results, we draw three key observations: **i)** HR@20 exhibits a clear upward trend as the interaction scale increases, confirming that sparsity significantly affects recommendation accuracy. **ii)** On YM, TaTriGR achieves 56.8% HR@20 in the sparsest group, outperforming MCCR and HMC by 0.8% and 1.3%, respectively. This demonstrates its superior ability to handle ‘cold-start’ users. On RB, TaTriGR improves HR@20 from 78.9% with MCCR and 79.4% with HMC to 80.5%, again showing advantages in sparse settings. **iii)** The performance gap further widens in denser groups, where TaTriGR reaches 86.9% HR@20 on RB compared with 85.3% and 85.7% for MCCR and HMC. This indicates that targeting-aware propagation and auxiliary objectives are effective at leveraging richer interactions.

4.3 Ablation Study (RQ2)

To analyze the contribution of each component in TaTriGR, we conduct ablation studies by replacing or removing specific modules. The following variants are evaluated:

- **w/o Targeting:** Replace the user–criterion distribution by a uniform vector $w_{uc} = 1/|C|$ for all $c \in C$.
- **w/o IPD:** Remove the IPD loss and eliminate the alignment between user embeddings and criterion-weighted ideal points.

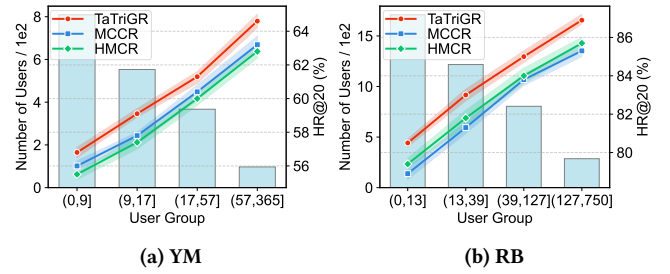


Figure 3: Performance comparison under different user interaction sparsity levels. Bars (left y-axis) show user counts, and lines (right y-axis) represent HR@20 values.

- **w/o LC:** Remove the LC loss and eliminate the ranking constraint on the user’s top-weighted criterion.
- **w/o IC:** Remove the item–criterion edges and keep only user–item and user–criterion connections.
- **w/o Criterion:** Remove all criterion-related edges and simplify TaTriGR into a user–item bipartite graph.

Table 3 presents the results of ablation studies, from which we derive the following insights: **i)** Every component is indispensable for enhancing TaTriGR. Performance consistently declines once any module is removed, indicating that targeting, auxiliary objectives, and item–criterion edges jointly contribute to the overall effectiveness of TaTriGR. **ii)** **w/o Targeting** leads to the most critical decline among the core components. Removing user–criterion targeting leads to a clear performance drop, highlighting the necessity of personalized criterion weights w_{uc} for capturing user preferences. Meanwhile, **w/o Criterion** causes the most severe

Table 3: Ablation studies of TaTriGR w.r.t. R@20 and N@20 on three datasets.

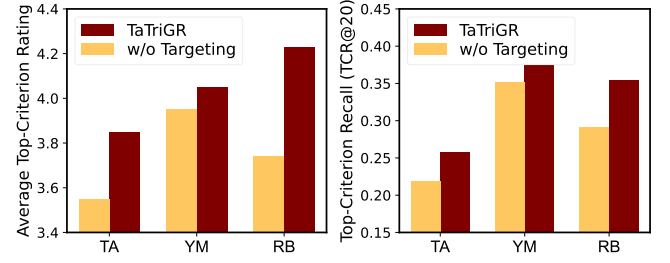
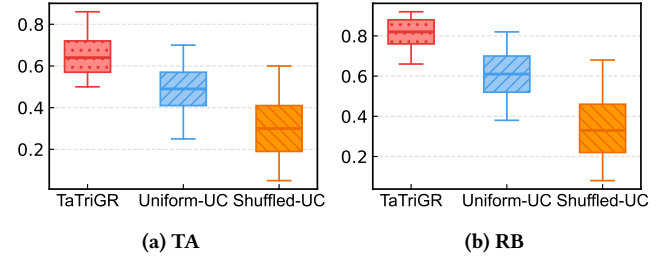
Model	TA		YM		RB	
	R@20	N@20	R@20	N@20	R@20	N@20
TaTriGR	0.1007	0.0981	0.2355	0.2942	0.4188	0.4293
<i>w/o Targeting</i>	0.0886	0.0857	0.2213	0.2790	0.3732	0.3869
<i>w/o IPD</i>	<u>0.0991</u>	<u>0.0962</u>	<u>0.2330</u>	<u>0.2910</u>	<u>0.4105</u>	<u>0.4212</u>
<i>w/o LC</i>	0.0968	0.0947	0.2319	0.2905	0.4057	0.4158
<i>w/o IC</i>	0.0901	0.0872	0.2238	0.2815	0.3824	0.3936
<i>w/o Criterion</i>	0.0654	0.0688	0.1757	0.1898	0.3074	0.2967

overall degradation, with R@20 decreasing by 35.1% on TA, 25.4% on YM, and 26.6% on RB, since eliminating all criterion signals degenerates TaTriGR into a single-rating recommender. **iii)** Both IPD and LC contribute positively, though with different emphases. **w/o IPD** shows the smallest impact among all ablations, with only moderate declines across datasets. This demonstrates that aligning users to criterion-weighted ideal points helps stabilize training and maintain reliable preference representations. In contrast, **w/o LC** mainly affects ranking quality, indicating that enforcing LC is beneficial when user preferences are dominated by a few key criteria. **iv)** **w/o IC** reveals the crucial role of item-side criterion information in TaTriGR. Removing this channel results in performance drops across all datasets. This confirms that item-criterion edges w_{ic} are essential for enriching item representations with criterion semantics. Without this channel, the model fails to capture how items perform under different criteria, causing weaker preference matching and poorer recommendation accuracy.

4.4 Targeting Analysis (RQ3)

To explore how user–criterion targeting influences recommendation performance and ranking behavior of TaTriGR, we analyze its contributions from multiple perspectives.

4.4.1 Top-Criterion Preferences. We examine how targeting aligns recommendation results with user-specific preferences from two perspectives. **i)** First, we measure the *Average Top-Criterion Rating*, defined as the mean rating of recommended items under each user’s dominant criterion c^* . Figure 4 shows that TaTriGR recommends items with higher ratings on c^* compared to **w/o Targeting**, with the greatest gains observed on RB, increasing from 3.15 to 3.52 on a 5-point scale. This indicates that targeting enables the model to better capture what users value most. **ii)** Second, we compute *Top-Criterion Recall (TCR@20)*, which evaluates how well the recommended list covers items relevant under the dominant criterion. As shown in Figure 4, TaTriGR substantially improves TCR over **w/o Targeting** on all three datasets, with the relative gains being largest on RB (+6.3%), moderate on TA (+3.9%), and smaller on YM (+2.2%). This pattern aligns with dataset characteristics, as the more heterogeneous the criteria are, the greater the benefit of targeting. Notably, YM exhibits the highest absolute TCR values because its criteria are highly correlated with overall ratings. Therefore, even without targeting, high-scoring items under overall ratings tend to align with top-criterion preferences.

**Figure 4: Analysis of targeting on top-criterion preferences.****Figure 5: Correlation between criterion-weighted aggregation and final scores on TA and RB datasets.**

4.4.2 Criterion-Weighted Aggregation. We verify whether TaTriGR leverages user–criterion targeting in the aggregation process by decomposing the predicted score into criterion-weighted and residual components, i.e., $\hat{y}_{ui} = g(u, i) + \psi(u, i)$, where $g(u, i) = \sum_c w_{uc} \phi_c(i)$. We then compute the correlation between $g(u, i)$ and $\hat{y}_{ui}$ across users, and compare TaTriGR with two baselines: *Uniform-UC*, which replaces w_{uc} with uniform weights, and *Shuffled-UC*, which randomly shuffles w_{uc} across users. As shown in Figure 5, TaTriGR achieves the highest and most concentrated correlation values on both TA and RB datasets. For example, on RB, the median correlation of TaTriGR is 0.82 with an interquartile range from 0.76 to 0.88, significantly higher than *Uniform-UC* with a median of 0.61 and *Shuffled-UC* with a median of 0.33. This indicates that criterion-weighted aggregation with targeting is well aligned with the final prediction scores. Meanwhile, *Shuffled-UC* exhibits the lowest and most unstable correlation, confirming that user-specific targeting is essential rather than a coincidental effect.

4.5 Sensitivity Analysis (RQ4)

To investigate the sensitivity of TaTriGR, we analyze the impact of four key hyperparameters on performance, including the weight λ_1 of loss $\mathcal{L}_{IPD}$, the weight λ_2 of loss $\mathcal{L}_{LC}$, the sharpness parameter α , and the smoothing parameter β . The results are shown in Figure 6.

4.5.1 Impact of λ_1 and λ_2 . As shown in Figure 6, both R@20 and N@20 values of TaTriGR steadily improve as λ_1 increases, reaching the best performance at $\lambda_1 = 1$. This confirms that aligning user embeddings with criterion-weighted ideal points helps stabilize training and enhances representation quality. In contrast, λ_2 has a more marked effect. When λ_2 is too small (e.g., $1e-4$ or $1e-3$), performance degrades significantly, which is consistent with the ablation results of removing LC. Performance improves as λ_2 increases, with

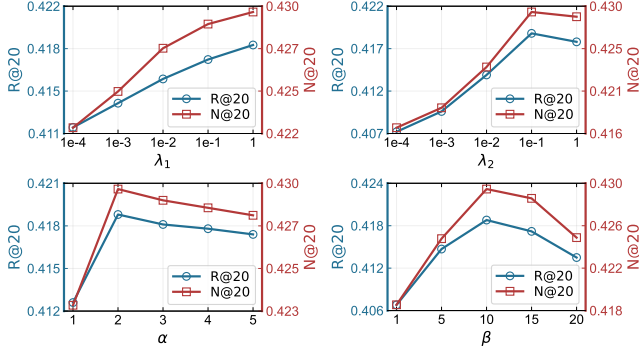


Figure 6: Hyperparameter sensitivities of λ_1 , λ_2 , α , and β w.r.t. $R@20$ and $N@20$ on the RB dataset.

the best results achieved at $\lambda_2 = 1e-1$. However, further increasing λ_2 to 1 leads to a decline, indicating that overly strong emphasis on LC may constrain the model’s flexibility.

4.5.2 Impact of α . The sharpness parameter α controls the concentration of the user–criterion weight distribution. As shown in Figure 6, performance first improves and then declines as α increases, achieving the best performance at $\alpha = 2$. This demonstrates that a moderate value allows TaTriGR to capture user-specific dominant criteria while still retaining auxiliary signals from other criteria. Too small a value (e.g., $\alpha = 1$) makes the distribution nearly uniform and weakens targeting mechanism, while too large a value (e.g., $\alpha = 5$) relies on a single criterion and degrades overall performance.

4.5.3 Impact of β . The smoothing parameter β balances item-level criterion scores with the global criterion mean. When $\beta = 10$, TaTriGR achieves the best performance on both $R@20$ and $N@20$, suggesting that it improves criterion estimates for items with few ratings while preserving discriminative signals for frequently rated items. However, when β is too small or too large, the model suffers from both noisy criterion estimates on sparse items and suppressed item-specific variations, which may cause inferior results.

5 Related Work

5.1 Multi-Criteria Recommendation

The core of multi-criteria recommendation is to enhance personalization by leveraging users’ ratings across multiple criteria instead of relying solely on overall ratings [6, 8, 31, 36]. Early approaches mainly focus on extending CF or MF techniques to multi-criteria settings [34, 37]. For example, MSVD [22] models user–item–criterion interactions as a tensor and applies multilinear singular value decomposition to capture latent relations. CLuCFEuc [24] clusters users into preference lattices based on their significant criteria and utilizes extended CF with Euclidean distance within each cluster. Later, deep learning methods are developed to capture nonlinear relationships among criteria. ExtendedSAE [37] modifies stacked autoencoders to incorporate multi-criteria ratings as input and predict their mapping to overall ratings. LatentMC [21] uses variational autoencoders to generate latent multi-criteria ratings from user reviews, which are then integrated with CF models. CFM [7] jointly

factorizes overall and multi-criteria rating matrices in an end-to-end latent factor framework. AEMC [34] employs two neural networks to first predict criterion-level ratings and then aggregate them into overall ratings. Although these methods enrich the modeling capacity of MCRSs, they generally treat criteria as additional information without explicitly modeling user preferences. To address this limitation, our proposed TaTriGR introduces user-specific criterion weights, enabling personalized targeting over multiple criteria.

5.2 Graph-based Recommender Systems

Graph-based recommender systems have emerged as a powerful paradigm by modeling user–item interactions as bipartite graphs and learning representations through message passing in graph neural networks (GNNs). Representative approaches include SpectralCF [45], NGCF [40], and LightGCN [13], which leverage high-order connectivity to refine user and item representations. Further advancements such as SGL [41] and AutoCF [42] leverage self-supervised learning and automated graph augmentation to handle sparsity and noise. Recently, several methods extend graph-based recommendation to multi-criteria settings. CPA-LGC [31] is the first model to introduce GNNs into MCRSs, providing a criterion-aware propagation strategy to integrate multiple criteria. CA-GF [32] proposes a training-free framework that constructs user-expansion graphs, applies criterion-specific graph filters, and aggregates user preferences for efficient recommendations. HMCR [11] adopts hyperbolic manifold modeling for MCRSs, mapping user–item representations into hyperbolic space and integrating multi-criteria ratings. MCCR [12] builds a structural causal model to mitigate bias in multi-criteria ratings and employs GNNs to capture heterogeneous user preferences. These approaches demonstrate the benefits of incorporating criteria signals into graph models. However, they often rely on static aggregation schemes and lack modeling of item–criterion channels. In contrast, our proposed TaTriGR introduces item–criterion channels within a tripartite graph and adaptive aggregation mechanisms to achieve more effective multi-criteria recommendations.

6 Conclusion

In this work, we propose **TaTriGR**, a targeting-aware tripartite graph recommender for multi-criteria rating. Its core idea is to model user–item–criterion interactions within a unified graph structure and incorporate user-specific criterion weights to capture heterogeneous preferences. Beyond the tripartite lightweight propagation, TaTriGR introduces two auxiliary objectives: IPD, which strengthens user representations, and LC, which enforces criterion-level prioritization. An additional item–criterion channel further enriches item embeddings with criterion semantics. Through score decomposition, targeting is made explicit and ensures fundamental decision properties including *Neutrality*, *Contribution*, *Responsiveness*, and *Contraction*. Experiments on three benchmark datasets demonstrate that TaTriGR consistently outperforms competitive baselines, validating the effectiveness of its targeting-aware design.

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A Target-Based Formalization

This section provides a theoretical formalization of target-based decision relations, which complements the main methodology by clarifying the semantics of user–criterion targeting. It also serves as a foundation for understanding the decision properties analyzed in Section 3.2.3.

Give an item set $I = \{i_1, i_2, \dots, i_{|I|}\}$ and a criterion set $C = \{c_1, c_2, \dots, c_{|C|}\}$, each criterion $c_m \in C$ is defined as a function $c_m : I \rightarrow \mathbb{R}_+$, assigning a non-negative evaluation to every item under the m -th dimension. For any item $i \in I$, the criterion evaluation vector is $\gamma(i) = (c_1(i), c_2(i), \dots, c_{|C|}(i))$, which corresponds to the criterion-level ratings in $\{\mathcal{R}_c\}_{c=1}^{|C|}$. A support set is a subset $S \subseteq C$ representing the criteria relevant in a given decision context. For a target item $t \in I$, its support is defined as $S(t) = \{c_m \in C \mid c_m(t) > 0\}$.

For two items $i, j \in I$ and a support set $S \subseteq C$, the dominance relations are defined as follows:

- $i \succeq_S j$: item i *Pareto Dominates* item j on S , meaning that i performs at least as well as j on every criterion in S .
- $i \succ_S j$: item i *Strictly Dominates* item j on S , meaning that $i \succeq_S j$ holds and $j \not\succeq_S i$; i.e., i is strictly better on at least one criterion without being worse on others.
- $i \simeq_S j$: item i and item j are *Equivalent* on S , meaning both $i \succeq_S j$ and $j \succeq_S i$ hold; i.e., they achieve identical evaluations on all criteria in S .

Given a candidate set $O \subseteq I$ and a target item t , the target-relative optimal set is defined as

$$O_t(S) = \{i \in O \mid \nexists j \in O, j \succ_{S(t)} i\}. \quad (21)$$

This set contains the items in O that are not strictly dominated with respect to the support criteria of the target t . A targeting function is a mapping $\beta : 2^I \times I \rightarrow 2^I$, such that for any candidate set $O \subseteq I$ and target item $t \in I$, $\beta_t(O) \subseteq O_t(S)$.

B Pseudocode of TaTriGR

In this section, we present the pseudocode of the proposed TaTriGR in Algorithm 1. The model starts by initializing user, item, and criterion embeddings, and constructs a targeting-aware tripartite graph with user–criterion and item–criterion edges weighted by personalized distributions. Lightweight propagation integrates collaborative signals and targeting semantics. Score decomposition is used for criterion-weighted aggregation, and two auxiliary objectives (IPD and LC) are incorporated. The overall loss jointly optimizes ranking and targeting objectives.

C Analysis of Corollaries

C.1 Proof of Theorem 3.1

Proof. By the decomposition from Eq. (14) together with assumption iii), we have for any $i \in O$,

$$\hat{y}_{ui} = \sum_{c \in C} w_{uc} \phi_c(i) + \psi_u. \quad (22)$$

Hence the ranking over the candidate set O satisfies

$$\arg \max_{i \in O} \hat{y}_{ui} = \arg \max_{i \in O} \sum_{c \in C} w_{uc} \phi_c(i) = \arg \max_{i \in O} f^{(u)}((\phi_c(i))_{c \in C}), \quad (23)$$

Algorithm 1 TaTriGR Algorithm

Input: User set $\mathcal{U}$, item set $\mathcal{I}$, criterion set $\mathcal{C}$; Multi-criteria rating data $\mathcal{D}$; Number of propagation layers L ; Embedding dimension d ; Learning rate η ; Loss hyperparameters $\lambda_1, \lambda_2, \lambda$; Sharpness parameter α ; Smoothing parameter β .

Output: Final user embeddings $\mathbf{e}_u^*$ and item embeddings $\mathbf{e}_i^*$.

- 1: Initialize $\mathbf{e}_u^{(0)}, \mathbf{e}_i^{(0)}, \mathbf{e}_c^{(0)}$ for all $u \in \mathcal{U}, i \in \mathcal{I}, c \in \mathcal{C}$.
- 2: Construct the tripartite graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with nodes $\mathcal{V} = \mathcal{U} \cup \mathcal{I} \cup \mathcal{C}$, where user–criterion weights w_{uc} are computed by Eq. (5) and item–criterion weights w_{ic} are derived from Eq. (9). The edge set $\mathcal{E}$ then includes user–item (u, i) links from interactions in $\mathcal{D}$, user–criterion (u, c) links with weights w_{uc} , and item–criterion (i, c) links with weights w_{ic} .
- 3: **for** each epoch **do**
- 4: **for** $l = 1$ **to** L **do**
- 5: Propagate embeddings with normalized adjacency $\hat{\mathbf{A}}$:
- 6: $\mathbf{e}_u^{(l+1)} = \text{ELU}(\sum_{i \in \mathcal{N}_u} \hat{\mathbf{A}}_{ui} \mathbf{e}_i^{(l)} + \sum_{c \in \mathcal{C}} \hat{\mathbf{A}}_{uc} \mathbf{e}_c^{(l)})$,
- 7: $\mathbf{e}_i^{(l+1)} = \text{ELU}(\sum_{u \in \mathcal{N}_i} \hat{\mathbf{A}}_{iu} \mathbf{e}_u^{(l)} + \sum_{c \in \mathcal{C}} \hat{\mathbf{A}}_{ic} \mathbf{e}_c^{(l)})$,
- 8: $\mathbf{e}_c^{(l+1)} = \text{ELU}(\sum_{u \in \mathcal{U}} \hat{\mathbf{A}}_{cu} \mathbf{e}_u^{(l)} + \sum_{i \in \mathcal{I}} \hat{\mathbf{A}}_{ci} \mathbf{e}_i^{(l)})$.
- 9: **end for**
- 10: Obtain final embeddings using Eq. (12).
- 11: Calculate predicted scores $\hat{y}_{ui} = \langle \mathbf{e}_u^*, \mathbf{e}_i^* \rangle$.
- 12: Decompose scores via Eq. (14).
- 13: Compute BPR loss $\mathcal{L}_{\text{BPR}}$ (Eq. (17)), IPD loss $\mathcal{L}_{\text{IPD}}$ (Eq. (18)), and LC loss $\mathcal{L}_{\text{LC}}$ (Eq. (19)).
- 14: Optimize the final objective $\mathcal{L}$ as defined in Eq. (20).
- 15: Update parameters with learning rate η .
- 16: **end for**
- 17: **return** user/item embeddings $\mathbf{e}_{u/i}^*$.

where $f^{(u)}(v_1, \dots, v_{|C|}) = \sum_c w_{uc} v_c$. It follows that the ranking depends only on $\{w_{uc}\}$ and $\{\phi_c(i)\}$, but not on item identities, which establishes **P1 (Neutrality)**. If $w_{uc} = 0$, the corresponding term vanishes from the sum, so criterion c has no effect, which gives **P2 (Contribution)**. For **P3 (Responsiveness)**, if $\phi_c(i)$ strictly increases for some c with $w_{uc} > 0$, then $\sum_c w_{uc} \phi_c(i)$ and consequently $\hat{y}_{ui}$ also strictly increase, ensuring that the ranking reflects this improvement. Finally, **P4 (Contraction)** holds because the aggregation $f^{(u)}$ is applied pointwise: restricting the candidate set from O to any subset $O' \subseteq O$ leaves the relative order of items in O' unchanged.

C.2 Corollary C.1

COROLLARY C.1. Consider the scoring function in Eq. (14) under the conditions of Theorem 3.1. Then these two properties hold: i) **Relevance**. If the weight w_{uc} of a criterion c increases while items differ only on c , then the preferred item for user u is the one with higher $\phi_c(i)$ value. In other words, raising the importance of c makes the ranking strictly more sensitive to improvements on c . ii) **Degenerate case**. If $w_{uc} = 1/|C|$ for all $c \in C$, the scoring function reduces to the uniform aggregation

$$f^{(u)}(v_1, \dots, v_{|C|}) = \frac{1}{|C|} \sum_{c=1}^{|C|} v_c, \quad (24)$$

and the propagation becomes equivalent to a non-targeted diffusion process.

Proof. i) For Relevance, from Eq. (14) and assumption iii) of Theorem 3.1, we have

$$\hat{y}_{ui} - \hat{y}_{uj} = \sum_{c'} w_{uc'} (\phi_{c'}(i) - \phi_{c'}(j)). \quad (25)$$

When i and j differ only on criterion c , the differences for all $c' \neq c$ vanish. The above expression thus reduces to

$$\hat{y}_{ui} - \hat{y}_{uj} = w_{uc} (\phi_c(i) - \phi_c(j)). \quad (26)$$

If $\phi_c(i) > \phi_c(j)$, then $\hat{y}_{ui} > \hat{y}_{uj}$ for any $w_{uc} > 0$. Moreover, increasing w_{uc} enlarges the difference $\hat{y}_{ui} - \hat{y}_{uj}$, making the ranking more sensitive to criterion c . Thus, raising the importance of c strictly reinforces the preference for items superior on c . ii) For the degenerate case, if $w_{uc} = 1/|C|$ for all c , then the aggregation reduces to the form of Eq. (24). In this case, the user-specific targeting distribution collapses to a uniform distribution over criteria. Hence the propagation no longer reflects personalized preferences and degenerates to a uniform, non-targeted diffusion process, which coincides with LightGCN [13] on the tripartite graph.

C.3 Connections to MCDM Rules

We further establish the relationship between the targeting formulation and two classical decision rules in MCDM: the Weighted Sum rule and the Lexicographic rule. These connections illustrate how targeting provides a unifying framework that subsumes traditional approaches under specific weighting schemes.

COROLLARY C.2. *If the user–criterion weights are uniform, i.e.,*

$$w_{uc} = \frac{1}{|C|}, \quad \forall c \in C, \quad (27)$$

then the targeting score reduces to the arithmetic mean of criterion evaluations:

$$\hat{y}_{ui} = \frac{1}{|C|} \sum_{c=1}^{|C|} \phi_c(i), \quad (28)$$

which is equivalent to the Weighted Sum rule with equal weights.

COROLLARY C.3. *If the targeting distribution degenerates to a single dominant criterion, i.e.,*

$$w_{uc} = \begin{cases} 1, & c = c^*, \\ 0, & c \neq c^*, \end{cases} \quad c^* \in C. \quad (29)$$

Then the preference reduces to $\hat{y}_{ui} = \phi_{c^}(i)$, which is equivalent to the Lexicographic rule that ranks items solely based on the dominant criterion c^* .*

D Additional Experimental Results

In this section, we present additional empirical experiments to further analyze the flexibility and effectiveness of the proposed TaTriGR framework.

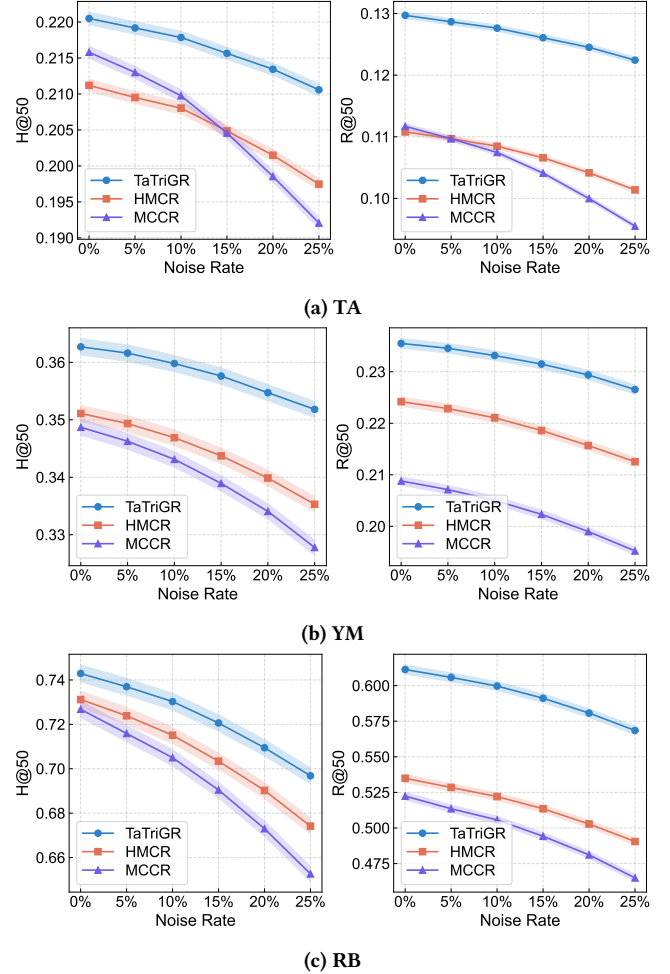


Figure 7: Performance comparison of TaTriGR, HMCR, and MCCR under varying noise rates. The shaded area represents the 95% confidence intervals based on a t -distribution over five random seeds.

D.1 Dataset Description

TripAdvisor (TA): The TA dataset comes from the TripAdvisor travel platform and provides hotel rating information. It includes an overall rating and seven criteria: *Business*, *Check-in Quality*, *Cleanliness*, *Location*, *Rooms*, *Service*, and *Value*. All ratings are on a scale of 1 to 5.

Yahoo! Movies (YM): The YM dataset comes from the Yahoo! Movies platform and contains movie rating information. It includes an overall rating and four criteria: *Story*, *Acting*, *Direction*, and *Visuals*. All ratings are on a scale of 1 to 5.

RateBeer (RB): The RB dataset comes from the RateBeer platform and provides beer rating information. It includes an overall rating and four criteria: *Appearance*, *Aroma*, *Taste*, and *Palate*. The ratings are on a scale of 1 to 5 (*Appearance* and *Palate*), 1 to 10 (*Aroma* and *Taste*), and 1 to 20 (overall).

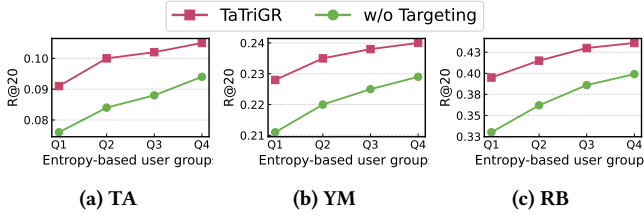


Figure 8: Comparison of TaTriGR and *w/o Targeting* in terms of R@20 across entropy-based user groups.

D.2 Noise Analysis

To evaluate the robustness of TaTriGR, we compare its performance with HMCR and MCCR under varying noise rates. Noise is simulated by injecting false positive interactions into the graph, with noise levels ranging from 0% to 25%. As shown in Figure 8, we have the following observations: **i)** Across all datasets and noise levels, TaTriGR consistently outperforms HMCR and MCCR, confirming the effectiveness of its targeting-aware design in mitigating the influence of noisy edges. **ii)** Performance degradation is smallest on YM, moderate on TA, and largest on RB. This aligns with dataset characteristics: YM is relatively denser and less sensitive to injected noise, while RB is highly sparse and involves heterogeneous rating scales, making it more vulnerable. **iii)** TaTriGR degrades more gracefully compared to the baselines. For example, on RB, when the noise rate increases from 0% to 25%, TaTriGR’s H@50 and R@50 decrease by about 6.2% and 7.0%, whereas HMCR drops by around 8.0%–9.0% and MCCR suffers the largest decline of over 10%.

D.3 Impact of Targeting across Entropy Levels

To investigate how targeting benefits users with different degrees of preference concentration, we divide users into four quartile groups based on the normalized entropy of their targeting distributions. Q1 corresponds to users with the sharpest preferences (low entropy), while Q4 represents users with the flattest preferences (high entropy). Figure 8 reports the results in terms of R@20 on the TA, YM, and RB datasets. Across all datasets, TaTriGR outperforms *w/o Targeting*, confirming the effectiveness of user-criterion targeting. **i)** On TA, the improvements are steady across groups, with a relatively uniform gap of about 0.010–0.015. This shows that personalized criterion weights capture subtle preferences even in relatively homogeneous domains. **ii)** On RB, the benefits of targeting are most evident for sharp-preference users in Q1, where TaTriGR improves R@20 by nearly 20%. This highlights the value of targeting under severe sparsity and heterogeneous criteria. **iii)** On YM, both models follow a similar increasing trend, and the gap between them remains small. This reflects that strongly correlated criteria provide only limited benefits from targeting, rather than dominant factor.

D.4 Impact of Hyperparameters L and d

We examine the impact of GNN layers L and the embedding dimension d on TaTriGR’s performance, since they directly influence representation quality and computational efficiency.

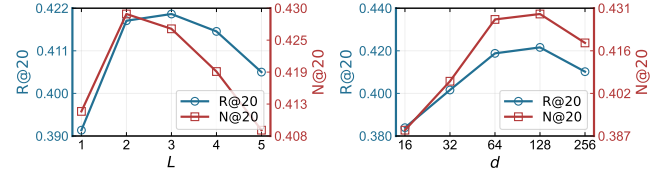


Figure 9: Hyperparameter sensitivities of L and d w.r.t. R@20 and N@20 on the RB dataset.

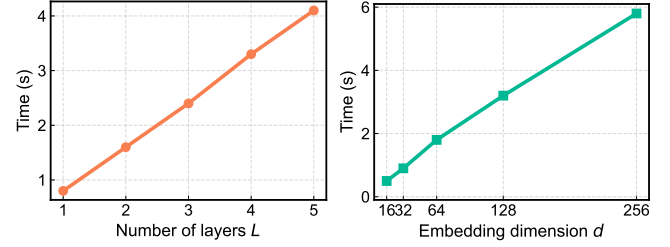


Figure 10: Computational complexity of TaTriGR w.r.t. the number of layers L and embedding dimension d .

D.4.1 Impact of L . As illustrated in Figure 9, increasing GNN layers L from 1 to 2 improves the performance of TaTriGR through richer information propagation. However, deeper models ($L \geq 3$) result in performance saturation or even minor drops due to over-smoothing, where node embeddings lose their discriminative power.

D.4.2 Impact of d . Increasing the embedding dimension d generally enhances model performance, as higher-dimensional embeddings provide richer representation capacity. However, as shown in Figure 9, performance peaks at $d = 128$ and slightly drops at $d = 256$, indicating that excessively large dimensions may lead to redundancy and overfitting while also incurring higher computational costs. Hence, a moderate setting such as $d = 64$ offers the best balance between accuracy and efficiency.

D.5 Computational Efficiency

We briefly analyze the computational complexity of TaTriGR. Let $|\mathcal{U}|$, $|\mathcal{I}|$, and $|\mathcal{C}|$ denote the numbers of users, items, and criteria, and let $|\mathcal{E}|$ be the total number of edges in the tripartite graph. The overall complexity can be summarized as follows. **i)** Constructing the tripartite graph requires $O(|\mathcal{E}|)$ operations, since edges are directly derived from observed interactions and smoothed criterion scores. **ii)** Each GNN layer aggregates messages along edges, with cost $O(|\mathcal{E}|d)$ per layer, where d is the embedding dimension. With L layers, this amounts to $O(L|\mathcal{E}|d)$. **iii)** Computing prediction scores $\hat{y}_{ui}$ and corresponding losses involves inner products and criterion-weighted decompositions, costing $O(d)$ per prediction and scaling linearly with the number of training samples. Therefore, the total training complexity of TaTriGR is $O(L|\mathcal{E}|d + |\mathcal{D}|d)$, which grows linearly with the graph size and embedding dimension. This explains the empirical results in Figure 10, where the training time grows nearly linearly with both L and d . The lightweight propagation design therefore enables manageable computational overhead as model depth or embedding dimensionality increases.